

FleetOps AI

An AI Operations Agent for Construction Equipment

Project Report

Track: AI + Construction Machinery → AI + Operation

15th China International College Students' Innovation & Entrepreneurship Competition

1. Executive Summary

Construction fleet managers currently rely on manual review of spreadsheets and periodic reports to track equipment performance — a slow process that delays the detection of downtime spikes, fuel waste, and maintenance issues. **FleetOps AI** is an AI Operations Agent that lets managers ask natural-language questions about their fleet (e.g. *“Which equipment has the highest downtime?”*) and receive instant, data-grounded answers, insights, and actionable recommendations.

Within a 10-day sprint, the team built a working prototype covering the full pipeline: data ingestion, KPI computation, an AI agent with tool-calling, an interactive dashboard, and cloud deployment — demonstrating the core value proposition end-to-end rather than a partial concept.

Metric	Value
Fleet size	30 units
Monitoring period	2026-06-01 - 2026-08-29
Overall utilization rate	95.2%
Total operating hours	15,863 hrs
Total downtime hours	793 hrs
Total fuel consumption	293,831 L

2. Problem Statement & Objectives

The Problem

Construction fleets generate large volumes of operational data (hours, downtime, fuel, maintenance) that is rarely analyzed in real time. Managers spend hours manually cross-referencing spreadsheets to answer basic operational questions, and often discover equipment issues only after they become costly.

Objectives

- Let managers query fleet data using natural language instead of manual spreadsheet analysis.
- Automatically surface equipment needing urgent attention (high downtime, abnormal maintenance frequency).
- Detect abnormal patterns such as sudden fuel-consumption spikes that may indicate mechanical issues.
- Generate ready-to-use monthly fleet performance reports.
- Deliver all of the above through a simple, accessible web dashboard.

3. Methodology & System Architecture

The system follows a five-stage pipeline, moving from raw fleet data to actionable operational decisions:

Stage	Function
Data Input	Daily equipment logs: operating hours, downtime, fuel consumption, maintenance events, project, location
Data Processing	Cleaning, feature engineering, and KPI calculation (Pandas)
AI Agent	LLM + tool-calling layer that interprets natural-language questions and selects the right analysis function
Visualization & UI	Streamlit dashboard: interactive charts, KPI cards, and a natural-language chat interface
Output	Answers to user questions, visual analysis, recommendations, and performance reports

Tool Library (AI Agent Functions)

- **get_top_downtime_equipment** — ranks equipment by downtime rate.
- **get_equipment_needing_attention** — flags equipment with abnormal downtime, maintenance frequency, or fuel use (compared fairly within its own equipment type).
- **explain_utilization_trend** — analyzes weekly utilization trends and identifies the likely driver of change.
- **analyze_fuel_consumption** — detects recent abnormal fuel-consumption spikes per unit.
- **generate_fleet_report** — compiles a full fleet performance summary.

Tech Stack

Layer	Tools
Core development	Python
Data analysis	Pandas, NumPy
AI / LLM	OpenAI API (Chat Completions with Function Calling)
Visualization	Plotly, Matplotlib
Web interface	Streamlit
Version control & deployment	GitHub, Streamlit Community Cloud

4. Data

Given the 10-day sprint, the team generated a realistic synthetic operations dataset (30 units, 6 equipment types, 5 active projects, 90 days of daily logs — 2,700 records) with deliberately embedded operational issues, so the AI agent's insight-detection could be validated against known ground truth before being pointed at real fleet data. The schema (equipment ID, type, project, operating hours, downtime hours, fuel consumption, maintenance flag) matches what a real telematics or ERP export would provide, so the pipeline can be pointed at live data with no code changes.

5. Results & Insights

Running the AI agent against the dataset surfaced the following validated insights, confirming the detection logic works correctly before deployment on real data:

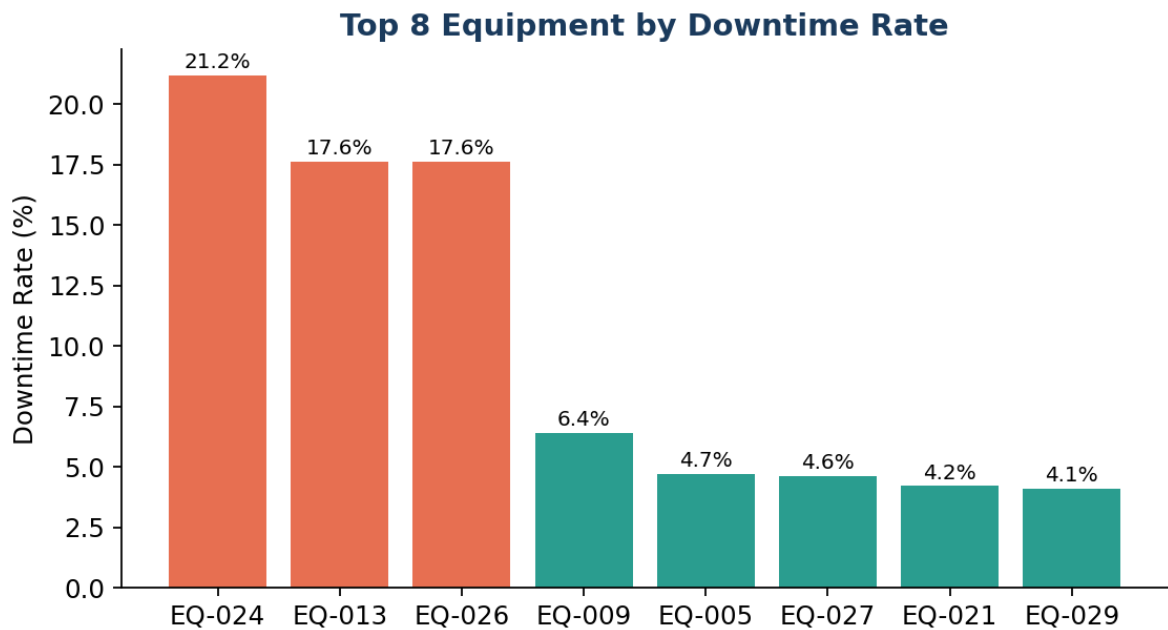


Figure 1 — Three units (EQ-024, EQ-013, EQ-026) show markedly higher downtime than the rest of the fleet, warranting priority inspection.

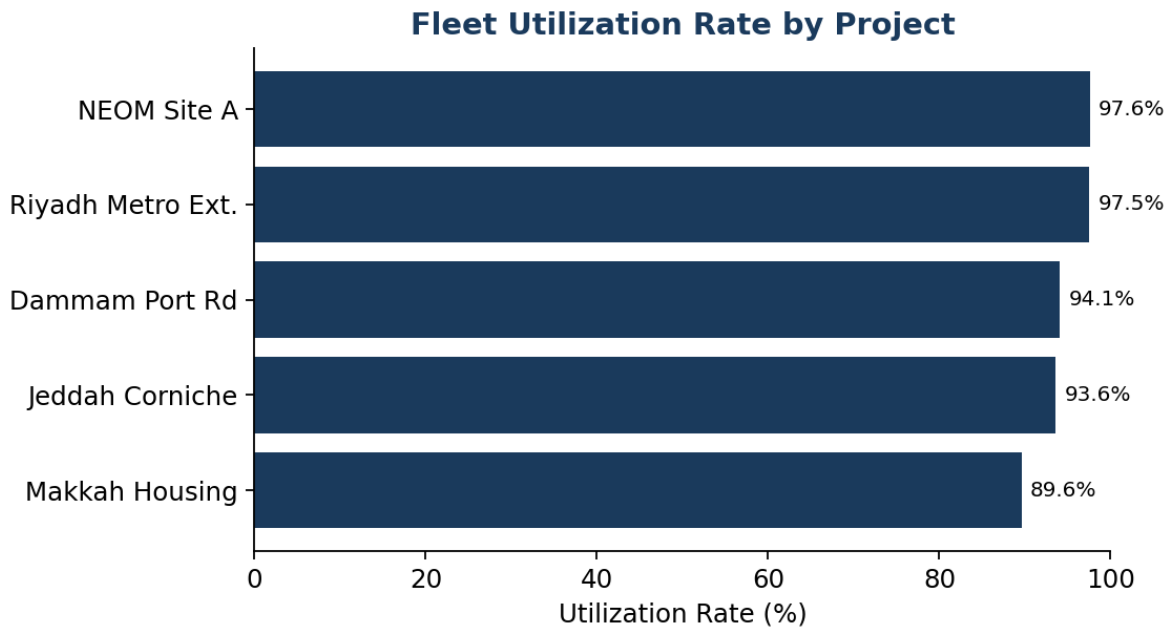


Figure 2 — Utilization rate by project. Makkah Housing shows the lowest utilization, correlating with its equipment's higher downtime.

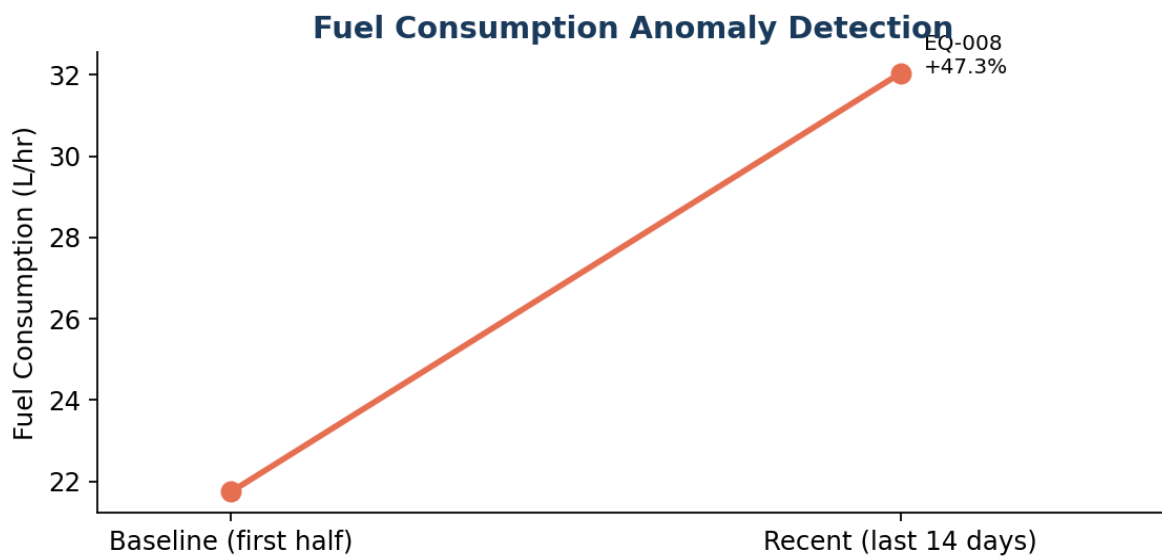


Figure 3 — Unit EQ-008 shows a 47% jump in fuel consumption per hour over the last 14 days versus its own baseline — a strong signal of a developing mechanical issue (e.g. leak, clogged filter) rather than simple increased usage.

Equipment Needing Immediate Attention

Equipment	Type	Project	Reason
EQ-024	Dump Truck	Dammam Port Rd	downtime ██████ (21.2%)
EQ-013	Grader	Makkah Housing	downtime ██████ (17.6%)
EQ-026	Dump Truck	Jeddah Corniche	downtime ██████ (17.6%)

EQ-012	Bulldozer	Makkah Housing	████████ ██████████ █████ ██████████ (3 ██████████ ██████████ 1.7)
EQ-014	Bulldozer	Riyadh Metro Ext.	████████ ██████████ █████ ██████████ (3 ██████████ ██████████ 1.7)
EQ-004	Excavator	NEOM Site A	████████ ██████████ █████ ██████████ (3 ██████████ ██████████ 1.7)

6. Actionable Recommendations

- **Inspect EQ-024, EQ-013, and EQ-026 immediately** — downtime rates of 17–21% are more than double the fleet average and are the single largest drag on overall utilization.
- **Schedule a mechanical check for EQ-008** — its sudden fuel-consumption increase is consistent with a leak or filter issue; addressing it now avoids further fuel waste and potential breakdown.
- **Review maintenance scheduling for EQ-012, EQ-014, and EQ-004** — their maintenance frequency is a statistical outlier versus the rest of the fleet, suggesting either a recurring fault or a scheduling inefficiency.
- **Investigate Makkah Housing's utilization decline** — it dropped from ~92% to ~84% over the monitoring period, likely tied to the downtime issues on its assigned equipment.

7. Project Status & Next Steps

Day	Milestone	Status
1–2	Project setup, data collection & exploration	Done
3	KPI calculation	Done
4	AI agent tool library	Done
5	LLM integration & agent logic	Done
6	Streamlit dashboard (overview, chat, reports)	Done
7	Advanced features (deeper anomaly detection)	Planned
8	Testing & refinement	Planned
9	Documentation & presentation	In progress (this report)
10	Final review & submission	Planned

Future Development

- Connect the agent to real telematics/ERP data sources (schema is already compatible).
- Add predictive maintenance (scikit-learn) to forecast failures before they occur, not just detect downtime after the fact.
- Extend the agent with proactive alerts (e.g. automatic Slack/email notification when a unit crosses a risk threshold).

8. Conclusion

FleetOps AI demonstrates that a small team can, within 10 days, build a working AI operations agent that turns raw construction-fleet data into natural-language insights and concrete recommendations. The

prototype validates the core value proposition end-to-end — from data to a decision a fleet manager can act on today — and is built on a data schema and architecture ready to scale to real operational data.